COMPLEX NUMBERS :->

Q1. Write the polar form of $3+4i$ and find its modulus and arguement.

Solution :-> We know that; $z = x+iy$

Polar form of $z = r(\cos\theta + \sin\theta)$

So, $z = 3+4i$

$\therefore \quad r = \sqrt{x^2+y^2} = \sqrt{3^2+4^2} = \sqrt{9+16}$

$\therefore \quad \boxed{r = 5} \qquad = \sqrt{25} = 5$

Now, Let, $\tan\alpha = \left|\frac{y}{x}\right|$

$= \left|\frac{4}{3}\right|$

$\Rightarrow \tan\alpha = \frac{4}{3}$

$\therefore$ The polar form $= 5\left(\cos\frac{4}{3} + \sin\frac{4}{3}\right)$.

Similarly the modulus $= 5$

And argument $= \frac{4}{3}$.

Q2. Simplify : $(2+i)^3$

Soln

$(2+i)^3$

$= (2)^3 + 3(2)^2(i) + 3(2)(i)^2 + (i)^3 \quad [(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3]$

$= 8 + 3.4i + 6(-1) + i^2.i$

$= 8 + 12i - 6 + (-1).i$

$= 8 + 12i - 6 - i$

$= 2 + 11i$

P.T.O.

Q3. Show that the cube root of unity are $1, \omega, \omega^2$ and that $\omega + \omega^2 + 1 = 0$.

Soln :-> we know that,

$$\omega = \frac{-1+i\sqrt{3}}{2} \quad ; \quad \omega^2 = \frac{-1-i\sqrt{3}}{2}$$

$$LHS = \frac{-1+i\sqrt{3}}{2} + \left(\frac{-1-i\sqrt{3}}{2}\right) + 1.$$

$$= \frac{-1+i\sqrt{3}}{2} - \frac{1-i\sqrt{3}}{2} + 1.$$

$$= \frac{-1-1+\cancel{i\sqrt{3}}-\cancel{i\sqrt{3}}}{2} + 1.$$

$$= \frac{-\cancel{2}}{\cancel{2}} + 1 \quad \cancel{= \frac{-2+}{2}}$$

$$= -1 + 1 = 0 \text{ —— RHS}$$

Q4. If $z = x + iy$ satisfies $|z| = 5$ and $\arg(z) = \pi/3$ find ~~x+y~~ x and y.

Solution -> Given,

$$|z| = r = 5.$$

$$\arg(z) = \theta = \frac{\pi}{3}$$

We, know that in polar form,

$$z = r(\cos\theta + i\sin\theta)$$

$$\Rightarrow z = 5\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$\because \cos\frac{\pi}{3} = \frac{1}{2} \quad ; \quad \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow z = 5\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$$

$$= \left(\frac{5}{2} + i\,\frac{5\sqrt{3}}{2}\right) \Rightarrow x = \frac{5}{2} \; ; \; y = \frac{5\sqrt{3}}{2}.$$

Q5. Devid
form a
Solution ->

PERMUTAT

Q6. In ho
be arran
Soln
Gi

Q5. Devide $4+2i$ by $1+i$ and express the reselt in the form $a+ib$.

Solution ->

$$\frac{4+2i}{1+i}$$

$$= \frac{4+2i}{1+i} \times \frac{1-i}{1-i}$$

$$= \frac{4(1) + 4(i) + 2i(1) + 2i(-i)}{1(1) + 1(-i) + i(1) + i(-i)}$$

$$= \frac{4 - 4i + 2i - 2(i)^2}{1 - i + i - (i)^2}$$

$$= \frac{4 - 4i + 2i - 2(-1)}{1 - \cancel{i} + \cancel{i} - (-1)}$$

$$= \frac{4 - 4i + 2i + 2}{1 + 1}$$

$$= \frac{4 - 2i + 2}{2} = \frac{\overset{3}{\cancel{6}} - 2i}{\cancel{2}}$$

$$= 3 - 2i.$$

Ans.

PERMUTATION AND COMBINATION.

Q6. In how many ways can the letters of the word ENGINE be arranged ?

Soln Given,

letters = E, N, G, I, N, E.

where, E occurs 2 times, N occurs 2 times.

$$\therefore \frac{n!}{n_1!\,n_2!\ldots} = \frac{6!}{2!\,2!} = \frac{\overset{180}{\cancel{720}}}{\cancel{2.4}} = 180$$

$\therefore$ these letter can be arranged in 180 different ways.

Q7. From a group of 8 men and 6 women, how many committee of 4 men and 3 women can be formed.

Soln

Let,

Choose 4 men out of 8 and 3 women out of 6. ~~use~~

$$\binom{8}{4} \times \binom{6}{3}$$

$$\frac{8!}{4!} = \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1} = \frac{1680}{24} = 70.$$

$$\frac{6!}{3!} = \frac{6 \times 5 \times 4}{3 \times 2 \times 1} = \frac{120}{6} = 20.$$

$$\Rightarrow 70 \times 20 = 1400$$

$\therefore$ 1400 different committee can be formed.

Q8. A student has to ~~co~~ choose 5 questions out of 8, such that at least 2 are form the first 4 and rest from the remaining 4. How many ways.

Soln

Given, A student need to select 5 questions of out of a total 8

The questions can be divided into.

First four questions (Group A) : 4 questions

Remaining forur questions (Group B) : 4 question.

$\therefore$ Possible combination to choose questions from group A is 2, 3 or 4.

Case : 1 -> If he ~~&~~ choose 2 Questions from A and 3 questions from B.

then, $C(4,2) = \frac{4!}{2!(4-2)!} = \frac{4 \times 3}{2 \times 1} = \frac{\cancel{12}^{6}}{\cancel{2}} = 6$

$C(4,3) = \frac{4!}{3!(4-3)!} = \frac{4}{1} = 4.$

$\therefore$ Total ~~co~~ ways $\Rightarrow 6 \times 4 = 24$

Case 2 :-> ~~it~~ If he choose 3 questions ~~fo~~ from A and 2 from B.

then, $C(4,3) = 4.$

$C(4,2) = 6.$

$\therefore$ Total ways $\Rightarrow 4 \times 6 = 24$

Case 3 :-> 4 questions from Group A and 1 questions from B

then, $C(4,4) = \frac{4!}{4!(4-4)!} = 1.$

$C(4,1) = \frac{4!}{1!(4-1)!} = 4.$

$\therefore$ total ways $= 4 \times 1 = 4.$

$\therefore$ Total number of ways he can choose questions is ~~24 x 24 =~~ $24 + 24 + 4 = 52$

Q9. How many ways can 5 distinct books be placed on a shelf if two specific books must always be together?

~~Ans~~ Solⁿ :-> Let,

two specific book be A and B.

$\because$ A & B must be together

$\therefore$ AB, Book 3, Book 4, Book 5 = 4 items

P.T.O

$\therefore \quad 4! = 4 \times 3 \times 2 \times 1 = 24$

$2! = 2 \times 1 = 2$

$\Rightarrow \quad 4! \times 2! = 24 \times 2$

$= 48$

$\therefore$ The books can be placed in 48 ways.

Q10. Find the number of different 4-digit even numbers that can be formed using digits 1, 2, 3, 4, 5 without repetition.

<u>Solution</u> -> Let,

The four digit number be P_1, P_2, P_3, P_4.

The digit that to be choose for P_4 are 2, 4

$\therefore \quad P(4,3) = \dfrac{4!}{(4-3)!} = \dfrac{4!}{1!} = 4 \times 3 \times 2 = 24.$

$\therefore$ The number of ways to arrange remaining two digits $= 24$.

Also,

The total number of four digit number that can be arranged

$\Rightarrow$ ~~dor~~ choices for P_4 × Permutation of P_1, P_2, P_3

$\Rightarrow \quad 2 \times 24 = 48$.

P.T.O.

BINO

Q11. E

Soln

BINOMIAL THEOREM.

Q11. Expand $(1+x)^5$ and find the coefficient of x^3.

Sol^n

We know that,

Binomial theorem $= (a+b)^n = \sum_{K=0}^{n} \binom{n}{K} a^{n-K} b^K$.

$\Rightarrow (1+x)^5 = \binom{5}{0} (1)^5 (x)^0 + \binom{5}{1} (1)^4 (x)^1 + \binom{5}{2} (1)^3 (x)^2$
$+ \binom{5}{0} (1)^2 (x)^3 + \binom{5}{0} (1)^1 (x)^4 + \binom{5}{5} (1)^0 (x)^5$

$\Rightarrow (1+x)^5 = 1(1) + 5(x) + 10(x^2) + 10(x^3) + 5(x^4) +$
$1(x^5)$

$\Rightarrow (1+x)^5 = 1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5$

Now,

To find the coefficient of x^3.

$\Rightarrow \quad T_{K+1} = \binom{n}{K} a^{n-k} \cdot b^K$.

$\Rightarrow \quad T_{3+1} = \binom{5}{3} (1)^{5-3} (x)^3$

$= \binom{5}{3} (1)^2$

$\binom{5}{3} = \frac{5!}{3!(5-3)!} = \frac{5!}{3!2!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{(3 \times 2 \times 1)(2 \times 1)}$

$= \frac{5 \times 4}{2} = \frac{\cancel{20}^{10}}{\cancel{2}}$.

$\therefore$ coefficient of $x^3 = 10$.

P.T.O →

Q12. Find the middle term in the expansion of $(x + 1/x)^8$.

Solution ->
$$\left(x + \frac{1}{x}\right)^8 \; ; n = 8.$$

$\Rightarrow n + 1 = 9^{th}$ term

$\because$ 9 is a odd number.

$\therefore$ middle term will be $\left(\frac{n}{2} + 1\right)^{th}$

$$= \frac{8}{2} + 1 = 4 + 1 = 5^{th}$$

Now,
$$T_{r+1} = \binom{n}{r} a^{n-r} b^r$$

$$\Rightarrow T_{4+1} = \binom{8}{4} (x)^{8-4} \left(\frac{1}{x}\right)^4$$

$$\Rightarrow T_5 = \binom{8}{4} (x)^4 \left(\frac{1}{x}\right)^4$$

$$\Rightarrow T_5 = \binom{8}{4} \cdot \cancel{x^4} \cdot \frac{1}{\cancel{x^4}} = \binom{8}{4}$$

$$\binom{8}{4} = \frac{8!}{4!(8-4)!}$$

$$= \frac{8!}{4!(4)!} = \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1} = \frac{1680}{24} = 70.$$

Q13. Expand $(2x - 3)^4$.

Solution -> We know that,
$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

$$\Rightarrow (2x + (-3))^4 = \binom{4}{0}(2x)^4(-3)^0 + \binom{4}{1}(2x)^3(-3)^1 + \binom{4}{2}(2x)^2 (-3)^2 + \binom{4}{3}(2x)^1(-3)^3 + \binom{4}{4}(2x)^0(-3)^4$$

$$= 1 \cdot (16x^4) \cdot 1 + 4 \cdot (8x^3) \cdot (-3) + 6 \cdot (4x^2)(9) + 4 \cdot (2x) \cdot (-27) + 1 \cdot 1 \cdot 81$$

$$= 16x^4 + (-96x^3) + 216x^2 + (-216x) + 81$$

$\Rightarrow (2x - 3)$

Q14) Using ...
three decim...

Solution ->

$(x+$

$\Rightarrow (1$

$\therefore$ ap...

Q15) Write t...
and hence...

Solution :->

$\Rightarrow (2x-3)^4 = 16x^4 - 96x^3 + 216x^2 - 216x + 81.$

Q14) Using the binomial theorem, approximate $(1+0.02)^{10}$ up to three decimal places.

Solution → We know that,

$$(x+y)^n = \sum_{K=0}^{n} \binom{n}{K} x^{n-K} y^K.$$

$$\Rightarrow (1+0.02)^{10} = \sum_{K=0}^{10} \binom{10}{K} (0.02)^K.$$

$$\Rightarrow \binom{10}{0}(0.02)^0 + \binom{10}{1}(0.02)^1 + \binom{10}{2}(0.02)^2$$
$$+ \binom{10}{3}(0.02)^3 + \binom{10}{4}(0.02)^4$$

$$\Rightarrow 1.1 + 10.(0.02) + 45.(0.0004) + 120.(0.000008)$$
$$+ 210.(0.00000016)$$

$$\Rightarrow 1 + 0.2 + 0.018 + 0.00096 + 0.0000336.$$

$$\Rightarrow 1.21896.$$

$\therefore$ approximate value of $(1+0.02)^{10} = 1.219$ approx.

Q15) Write the general term in the expansion of $(a+b)^n$ and hence find the term independent of x in $(2x + 3/x)^7$.

Solution :→ Here, $a = 2x$; $b = \frac{3}{x}$; $n = 7$.

$$\therefore T_{r+1} = \binom{7}{r} (2x)^{7-r} \left(\frac{3}{x}\right)^r.$$

$$= \binom{7}{r} . 2^{7-r} . x^{7-r} . 3^r . \left(\frac{1}{x}\right)^r.$$

$$= \frac{7}{r} . (2^{7-r} \cdot 3^r) . x^{7-r} . x^{-r}$$

$$= \frac{7}{r} (2^{7-r} 3^r) x^{(7-r)+(-r)}$$

$$= \frac{7}{r} (2^{7-r} 3^r) x^{7-2r}.$$

$\therefore \quad 7 - 2r = 0$

$2r = 7$

$r = \frac{7}{2} \Rightarrow r = 3.5$

$\because$ r must be an integer and ~~the~~ here $r = 3.5$

$\therefore$ There is no term independent to x.

~~Q16.~~ LOGARITHMMS

Q16. Evaluate $\log_2 8 + \log_2 16 - \log_2 4$.

Solution :->

$$\log_2 8 + \log_2 16 - \log_2 4$$

$$\Rightarrow 3 + 4 - 2 = 7 - 2 = 5$$

Q17. Prove that $\log ab \times \log_b a = 1$.

Solution -> LHS,

$$\log_a b \times \log_b a$$

$$\Rightarrow \frac{\log b}{\log a} \times \frac{\log a}{\log b} = 1 \quad \text{— RHS.}$$

$$* \left(\log_x y = \frac{\log_K y}{\log_K x} \right.$$

Q18. Solve for $x : 2^{(x-1)} = 8^{(2x+1)}$.

Solution ->

$$2^{(x-1)} = 2^{3(2x+1)}$$

$$\Rightarrow \cancel{2^{3(2x+1)}} \quad \cancel{2f} \quad 2^{(x-1)} = 2^{6x+3}$$

$$\Rightarrow -1 = 6x - x + 3$$

$$\Rightarrow -1 = 5x + 3$$

$$\Rightarrow -1 - 3 = 5x \quad \Rightarrow -4 = 5x. \Rightarrow x = \frac{-4}{5}$$

Q19. Express $\log_3 5$ in terms of natural logarithms.

Solution → we know that,

$$\log_b a = \frac{\log_K a}{\log_K b}$$

$$\Rightarrow \log_3 5 = \frac{\log_e 5}{\log_e 3}$$

$$\Rightarrow \log_3 5 = \frac{\ln 5}{\ln 3}$$

Ans

Q20. If $\log_x 81 = 4$, find x.

Soln :-> We know,

$$\log_b a = c$$

$$\Rightarrow b^c = a.$$

$$\Rightarrow x^4 = 81 \Rightarrow x = \sqrt[4]{81}.$$

$$\Rightarrow x = 3.$$

Ans.

SERIES (A.P & G.P)

Q21. Write the first 5 terms of an AP whose first term is 7 and common difference is 3.

Solution :-> We know,

$$a_n = a_1 + (n-1)d,$$

Given, $a = 7$; $d = 3$.

$$\therefore a_2 = 7+(2-1)3 = 7+3 = 10.$$

$$\Rightarrow a_3 = 7+(3-1)3 = 7+6 = 13$$

$$\Rightarrow a_4 = 7+(4-1)3 = 7+9 = 16$$

$$\Rightarrow a_5 = 7+(5-1)3 = 7+12 = 19.$$

∴ first 5 terms of A.P are -

7, 10, 13, 16, 19

Q22) Find the sum of the first 12 terms of the A.P : 5, 8, 11...

Solⁿ Given,

A.P = 5, 8, 11, ...

$\Rightarrow a_1 = 5$

$\Rightarrow d = a_2 - a_1 = 8 - 5 = 3$

$\Rightarrow n = 12$.

$\therefore S_n = \frac{n}{2}\left[2a_1 + (n-1)d\right]$

$\Rightarrow S_{12} = \frac{12}{2}\left[2.5 + (12-1)3\right]$

$\Rightarrow S_{12} = \frac{12}{2}\left[10 + 11.3\right] \Rightarrow S_{12} = \frac{\cancel{12}^{6}}{\cancel{2}}\left[10 + 33\right]$

$\Rightarrow S_{12} = 6\left[43\right] = 258$

$\therefore$ Sum of first 12 terms of the AP is $\boxed{258}$

Q23) Evalute the sum of the G.P : 16 + 8 + 4 + ... upto infinity.

Solⁿ Given,

G.P = 16 + 8 + 4 + ... ∞

$a_1 = 16$.

Common ratio $= \frac{8}{16} = \frac{1}{2}$

$\therefore S_\infty = \frac{a_1}{1-r} \Rightarrow S_\infty = \frac{16}{1-\frac{1}{2}}$

$\Rightarrow S_\infty = \frac{16}{\frac{1}{2}} \Rightarrow S_\infty = 16 \times \frac{2}{1}$.

$\Rightarrow S_\infty = 16 \times 2 = 32$ #

$\therefore$ The sum of the G.P ~~upto~~ 16 + 8 + 4 ... upto infinity is $\boxed{32}$

Q24. If th... vely, f...

Solⁿ :->

Now

Q25. For ... 1/2 fin...

Solⁿ Gi...

Q24. If the 5th and 8th terms of an A.P are 20 and 26 respectively, find 1st term and common difference.

Soln :-> Given,

$a_5 = 20$

$a_8 = 26$

we know that, $a_n = a_1 + (n-1)d$.

$\Rightarrow a_5 = a_1 + (5-1)d \Rightarrow a_1 + 4d = 20$ —— (I)

$\Rightarrow a_8 = a_1 + (8-1)d \Rightarrow a_1 + 7d = 26$ —— (II)

Now, by substracting eq 2 - eq 1

$$\begin{array}{r} \cancel{a_1} + 7d = 26 \\ \cancel{a_1} \underset{(-)}{+} 4d \underset{(-)}{=} 20 \\ \hline 3d = 6 \end{array}$$

$\Rightarrow d = \cancel{\frac{6}{3}}^{2} \Rightarrow d = 2.$

Now, putting $d = 2$ in in eq - (I) we get,

$a_1 + 4(2) = 20.$

$\Rightarrow a_1 + 8 = 20 \Rightarrow a_1 = 20 - 8$

$\Rightarrow a_1 = 12$

$\therefore$ First term is 12;

common difference is 2.

Q25. For a G.P with first term 2 and common ratio 1/2 find the sum of the first 8 term.

Soln Given,

$a_1 = 2$, $n = 8$.

$r = \frac{1}{2}$

$$\Rightarrow S_n = \frac{a_1(1-r^n)}{1-r}$$

$$\Rightarrow S_8 = \frac{2\left(1-\left(\frac{1}{2}\right)^8\right)}{1-\frac{1}{2}}$$

$$\Rightarrow S_8 = \frac{2\left(1-\frac{1^8}{2^8}\right)}{\frac{1}{2}}$$

$$\Rightarrow S_8 = \frac{2\left(1-\frac{1}{256}\right)}{\frac{1}{2}}$$

$$\Rightarrow S_8 = \frac{2\left(\frac{256-1}{256}\right)}{\frac{1}{2}}$$

$$\Rightarrow S_8 = \frac{2\left(\frac{255}{256}\right)}{\frac{1}{2}} \quad \Rightarrow S_8 = 2\left(\frac{255}{256}\right) \times \frac{2}{1}$$

$$= \frac{512}{256} \times 2$$

$$= \frac{1024}{256}.$$

Matrices

Q26. Define a square matrix and a zero matrix.

Solution-> A square matrix is $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}_{2\times 2}$.

And,

A zero matrix is $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Q27. Let $A = [1\ 3\ 2\ 4]$ find its transpose.

Soln

$$A^T = \begin{bmatrix} 1 \\ 3 \\ 2 \\ 4 \end{bmatrix}$$

Q28. If $A = [2\ -1\ 1\ 3]$ and $B = [0\ 2\ 5\ 1]$ compute $A+B$ and AB,

Soln

Here, $A = [2\ -1\ 1\ 3]$

$B = [0\ 2\ 5\ 1]$

$\therefore A+B = [2\ -1\ 1\ 3]\ [0\ 2\ 5\ 1]$

$= [2+0\quad -1+2\quad 1+5\quad 3+1]$

$= [2\quad 1\quad 6\quad 4]$ #

And, $AB = [2\ -1\ 1\ 3]\ [0\ 2\ 5\ 1]$

$= [2\times0\quad -1\times2\quad 1\times5\quad 3\times1]$

$= [0\quad -2\quad 5\quad 3]$ #

Q29. Show by example that $AB \neq BA$ in general for 2×2 matrices.

Soln

Let,

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \quad ; \quad B = \begin{bmatrix} 0 & 2 \\ 5 & 1 \end{bmatrix}$$

$$\therefore AB = \begin{bmatrix} (2)(0) + (-1)(5) & (2)(2) + (-1)(1) \\ (1)(0) + (3)(5) & (1)(2) + (1)(1) \end{bmatrix}$$

$$= \begin{bmatrix} 0 - 5 & 4 - 1 \\ 0 + 15 & 2 + 1 \end{bmatrix} = \begin{bmatrix} -5 & 3 \\ 15 & 3 \end{bmatrix}$$

//

Similarly, $BA = \begin{bmatrix} 0 & 2 \\ 5 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$

$$= \begin{bmatrix} 0+2 & -1+6 \\ 10+1 & -5+3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 5 \\ 11 & -2 \end{bmatrix}$$

$\therefore \quad AB \neq BA$ — Hence proved.

Q30. Solve the system using matrix method. $x+y+z=6$; $2x-y+3z=14$; $x+4y-z=2$.

Soln Here,

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 1 & 4 & -1 \end{bmatrix} ; \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} ; \quad B = \begin{bmatrix} 6 \\ 14 \\ 2 \end{bmatrix}$$

$$D = \det(A) = 1\begin{vmatrix} -1 & 3 \\ 4 & -1 \end{vmatrix} - 1\begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} + 1\begin{vmatrix} 2 & -1 \\ 1 & 4 \end{vmatrix}.$$

$$= 1\left((-1\cdot -1) - (3\cdot 4)\right) - 1\left((2)(-1) - (3)(1)\right) + 1\left((2\cdot 4) - (-1\cdot 1)\right)$$

$$= 1(1-12) - 1(-2-3) + 1(8-(-1))$$

$$= 1(-11) - 1(-5) + 1(9)$$

$$= -11 + 5 + 9$$

$$\Rightarrow -11 + 14 = 3$$

$$x = \begin{vmatrix} 6 & 1 & 1 \\ 14 & -1 & 3 \\ 2 & 4 & -1 \end{vmatrix}$$

$$= 6(1-12) - 1(-14-6) + 1(56-(-2))$$

P.T.O.

$= 6(-11)$

$= -66$

~~Dy =~~

$z =$

$\therefore$

DETER

Q31. C

Soln

$$= 6(-11) - 1(-20) + 1(58)$$
$$= -66 + 20 + 58 = 12.$$

~~Dy =~~ $y = \begin{vmatrix} 1 & 6 & 1 \\ 2 & 14 & 3 \\ 1 & 2 & -1 \end{vmatrix}$

$$= 1(-14-6) - 6(-2-3) + 1(4-14)$$
$$= 1(-20) - 6(-5) + 1(-10)$$
$$= -20 + 30 - 10 = 0.$$

$$z = \begin{vmatrix} 1 & 1 & 6 \\ 2 & -1 & 14 \\ 1 & 4 & 2 \end{vmatrix}$$

$$= 1(-2-56) - 1(4-14) + 6(8-(-1))$$
$$= 1(-58) - 1(-10) + 6(9)$$
$$= -58 + 10 + 54 = 6.$$

$$\therefore \quad x = \frac{X}{D} = \frac{12}{3} = 4$$
$$y = \frac{Y}{D} = \frac{0}{3} = 0.$$
$$z = \frac{Z}{D} = \frac{6}{3} = 2.$$

DETERMINENT.

Q31. Compute $\cancel{|1\,2\,3\,4|}$. $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$

Soln

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = (1)(4) - (2)(3)$$
$$= 4 - 6 = -2$$

Q32. Define the minor and Co-factor of an element in a 3×3 determinent.

Ans → For an element a_{ij} located in the ith row and jth column of a 3×3 determinent.

The minor of the element a_{ij} is the determinent of the 2×2 sub matrix obtained by deleting the ith row and jth column that contain a_{ij}.

The cofactor of the element a_{ij} is the minor M_{ij} multiplied by a sign factor, determined by the position of the element.

$$C_{ij} = (-1)^{i+j} M_{ij}$$

Q33. Evaluate. $\begin{vmatrix} 2 & 3 & 1 \\ 4 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix}$

Soln

$$A = \begin{vmatrix} 2 & 3 & 1 \\ 4 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix}$$

$$A = 2\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} - 3\begin{vmatrix} 4 & 2 \\ 1 & 3 \end{vmatrix} + 1\begin{vmatrix} 4 & 1 \\ 1 & 2 \end{vmatrix}$$

$$= 2(1\times3 - 2\times2) - 3(4\times3 - 2\times1) + 1(4\times2 - 1\times1)$$

$$= 2(3-4) - 3(12-2) + 1(8-1)$$

$$= 2(-1) - 3(10) + 1(7)$$

$$= -2 - 30 + 7 = -32 + 7 = -25$$

Q34. Solve using Camer's rule : $2x + 3y = 8$, $x - y = 2$

P.T.O

element in a
the ith row
determinent
g the ith
e minor M_{ij}
the position
$2 - 1 \times 1)$
5
$-y = 2.$
P.T.O

Soln

Here, $2x + 3y = 8$

$x - y = 2$

$\Rightarrow A = \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}$; $X = \begin{bmatrix} x \\ y \end{bmatrix}$; $B = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$

Let,

$|A| = \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix}$

$= (2)(-1) - (3)(1) = -2 - 3 = -5 = D$

Similarly $|X| = \begin{vmatrix} 8 & 3 \\ 2 & -1 \end{vmatrix}$

$= (8)(-1) - (3)(2) = -8 - 6 = -14$

$\Rightarrow |Y| = \begin{vmatrix} 2 & 8 \\ 1 & 2 \end{vmatrix}$

$\Rightarrow (2)(2) - (8)(1) = 4 - 8 = -4.$

$x = \frac{Dx}{D} = \frac{\cancel{-}14}{\cancel{-}5} = \frac{14}{5}$

$y = \frac{Dy}{D} = \frac{\cancel{-}4}{\cancel{-}5} = \frac{4}{5}$

$\therefore x = \frac{14}{5}$; $y = \frac{4}{5}$.

Q35. Prove that, $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a).$

Soln

LHS, $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$

$C_1 \to C_1 - C_2$, $C_2 \to C_2 - C_3$

$$= \begin{vmatrix} 1-1 & 1-1 & 1 \\ a-b & b-c & c \\ a^2-b^2 & b^2-c^2 & c^2 \end{vmatrix}$$

$$= \begin{vmatrix} a(a-b) & 0(b-c) & 1 \\ 1(a-b) & 1(b-c) & c \\ (a+b)(a-b) & (b+c)(b-c) & c^2 \end{vmatrix}$$

$$= (a-b)(b-c) \begin{vmatrix} 0 & 0 & 1 \\ 1 & 1 & c \\ a+b & b+c & c^2 \end{vmatrix}$$

$$= (a-b)(b-c) \left[a \begin{vmatrix} 1 & c \\ b\times c & c^2 \end{vmatrix} - 0 + 1 \begin{vmatrix} 1 & 1 \\ a+b & b+c \end{vmatrix} \right]$$

$$= (a-b)(b-c)\{0 + (b+c = a-b)\}$$

$$= (a-b)(b-c)(c-a) \text{ — RHS.}$$

TRIGONOMETRIC RATIOS

Q36. Prove $\sin^2 A + \cos^2 A = 1$

Soln ~~Given~~, LHS,

$$\sin^2 A + \cos^2 A.$$

$$= \left(\frac{y}{r}\right)^2 + \left(\frac{x}{r}\right)^2$$

$$= \frac{y^2}{r^2} + \frac{x^2}{r^2}$$

$$= \frac{x^2+y^2}{r^2} = \frac{\cancel{r^2}}{\cancel{r^2}} = 1 \text{ — RHS}$$

Q37. Express

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Let, A

Sin (x + y

=> Sin

Now,

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Q38. If ta

Soln

le

Q37. Express $\sin A + \sin B$ as
$2\sin[(A+B)/2)\cos((A-B)/2)$.

Solⁿ we know,

$$\sin(x+y) = \sin x \cos y + \cos x \sin y.$$

$$\sin(x-y) = \sin x \cos y - \cos x \sin y.$$

Let, $A = x + y$; $B = x - y$

$$\sin(x+y) + \sin(x-y) = (\sin x \cos y + \cos x \sin y) + (\sin x \cos y - \cos x \sin y)$$

$$\Rightarrow \sin A + \sin B = 2\sin x \cos y.$$

Now,

Expressing x & y in terms of A and B :->

$$A + B = (x+y) + (x-y) = 2x$$

$$x = \frac{A+B}{2}$$

$$A - B = (x+y) - (x-y) = 2y$$

$$y = \frac{A-B}{2}$$

$$\therefore \sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

Q38. If $\tan A = 1/2$ find $\sin 2A$ and $\cos 2A$,

Solⁿ we know that,

$$\sin 2A = \frac{2\tan A}{1+\tan^2 A}.$$

$$\cos 2A = \frac{1-\tan^2 A}{1+\tan^2 A}.$$

$$\Rightarrow \sin 2A = \frac{2\left(\frac{1}{2}\right)}{1+\left(\frac{1}{2}\right)^2} = \frac{1}{1+1/4}$$

$$= \frac{1}{5/4} = \frac{4}{5} \#$$

$$\Rightarrow \cos 2A = \frac{1-(1/2)^2}{1+(1/2)^2} = \frac{1-1/4}{1+1/4} = \frac{\frac{3}{\cancel{4}}}{5/\cancel{4}}$$

$$= \frac{3}{5} \#$$

Q 39. Prove that $\cos 3A = 4\cos^3 A - 3\cos A$.

Solⁿ LHS,

$$\cos 3A = \cos(2A + A)$$

$$= \cos 2A \cos A - \sin 2A \sin A$$

$$= \cancel{(2\cos^2}\,(2\cos^2 A - 1)\cos A - (2\sin A \cos A)\sin A$$

$$= 2\cos^3 A - \cos A - 2\sin^2 A \cos A.$$

$$= 2\cos^3 A - \cos A - 2(1-\cos^2 A)\cos A$$

$$= 2\cos^3 A - \cos A - 2\cos A + 2\cos^3 A$$

$$= (2\cos^3 A + 2\cos^3 A) + (-\cos A - 2\cos A)$$

$$= 4\cos^3 A - 3\cos A \text{ —— RHS.}$$

Q40. Show that $\sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B$

Sol

LHS,

$$\sin(A+B)\sin(A-B)$$

$$= (\sin A \cos B)^2 - (\cos A \sin B)^2$$

$$= \sin^2 A \cos^2 B - \cos^2 A \sin^2 B$$

$$= \sin^2 A(1-\sin^2 B) - (1-\sin^2 A)\cdot\sin^2 B$$

P.T.O

$= (\sin^2 A - \sin^2 A \sin^2 B) - (\sin^2 B - \sin^2 A \sin^2 B)$

$= \sin^2 A - \cancel{\sin^2 A \sin^2 B} - \sin^2 B + \cancel{\sin^2 A \sin^2 B}$

$= \sin^2 A - \sin^2 B$ — RHS.

~~Q4~~ INVERSE TRIGONOMETRIC FUNCTIONS ->.

Q41. Find $\sin^{-1}\left(\frac{1}{2}\right)$

Soln

Since,

$\sin\left(\frac{\pi}{6}\right) = 1/2$.

$\therefore \sin^{-1}(1/2) = \frac{\pi}{6}$ or $30°$

Q42. Prove that $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$

Soln

Let, $\theta = \sin^{-1} x$.

$\Rightarrow \sin\theta = x$.

$\Rightarrow x = \cos(\pi/2 - \theta)$ $\left[\sin\theta = \cos(\pi/2 - \theta)\right]$

$\Rightarrow \cos^{-1} x = \frac{\pi}{2} - \theta$

$\Rightarrow \cos^{-1} x = \pi/2 - \sin^{-1} x$.

$\Rightarrow \sin^{-1} x + \cos^{-1} x = \pi/2$.

← RHS.

Q43. Simplify. $\tan^{-1}(x) + \tan^{-1}(1/x)$ · for $x > 0$.

Soln

$\tan^{-1}(1/x) = \cot^{-1}(x)$

$\tan^{-1}(x) + \tan^{-1}(1/x) = \tan^{-1}(x) + \cot^{-1}(x)$

$\tan^{-1}(x) + \cot(x) = \pi/2$.

$\Rightarrow \tan^{-1}(x) + \tan^{-1}(1/x) = \pi/2$.

Q44. Find $\sin(\tan^{-1}(3/4))$.

Solⁿ

Let,

$\theta = \tan^{-1}(3/4)$

$\tan\theta = 3/4$

$\because \tan\theta = \dfrac{\text{Opposite}}{\text{Adjacent}} = \dfrac{3}{4}$

$\therefore$ By Pythogoras theorem,

$\text{Hypotenuse} = \sqrt{(\text{opposite})^2 + (\text{Adjacent})^2}$

$= \sqrt{3^2 + 4^2} = \sqrt{9+16}$

$= \sqrt{25} = 5.$

$\sin\theta = \dfrac{\text{Opposite}}{\text{Hypotenuse}} = \dfrac{3}{5}$

$\therefore \sin(\tan^{-1}(3/4)) = 3/5.$

Q45. Evaluate $\tan^{-1}(1) + \tan^{-1}(2) + \tan^{-1}(3)$

Solⁿ

$\tan^{-1}(1) = \pi/4.$

~~Here~~, we know, $\tan^{-1}x + \tan^{-1}y$ is $\tan^{-1}\left(\dfrac{x+y}{1-xy}\right)$

Here, $x = 2$; $xy = 6.$
$y = 3$

$\Rightarrow \tan^{-1}(2) + \tan^{-1}(3) = \pi + \tan^{-1}\left(\dfrac{2+3}{1-(2)(3)}\right)$

$= \pi + \tan^{-1}\left(\dfrac{5}{1-6}\right)$

$= \pi + \tan^{-1}(-1)$

$\tan^{-1}(2) + \tan^{-1}(3) = \pi + (-\pi/4) = 3\pi/4$

$\tan^{-1}(1) + \left(\tan^{-1}(2) + \tan^{-1}(3)\right) = \pi/4 + 3\pi/4$

$\Rightarrow \dfrac{4\pi}{4} = \pi$

PROPERTIES OF TRIANGLE

Q46. State and write the sine rule.

Ans-> The sine rule states that the ratio of the length of a side of a triangle to the sine of its opposite angle is constant for all three sides.

For any ΔABC,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

Q47. Derived the cosine rule for side a: $a^2 = b^2 + c^2 - 2bc\cos A$.

Soln

$$a^2 = (x_C - x_B)^2 + (y_C - y_B)^2$$

$$\Rightarrow a^2 = (b\cos A - c)^2 + (b\sin A - 0)^2$$

$$\Rightarrow a^2 = (b^2\cos^2 A - 2bc\cos A + c^2) + b^2\sin^2 A$$

$$\Rightarrow a^2 = b^2(\cos^2 A + \sin^2 A) + \cancel{\cos}\, c^2 - 2bc\cos A$$

$$\Rightarrow a^2 = b^2(1) + c^2 - 2bc\cos A$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

Q48. In ΔABC if $A = 30°$, $B = 45°$, $c = 10$ cm find a

Soln we know,

Sum of all angle of a Δ is $180°$

$$\therefore\ C = 180° - (A + B)$$

$$= 180 - (30 + 45) = 180 - 75$$

$$= 105°$$

$$\frac{a}{\sin A} = \frac{c}{\sin C}$$

$$a = c \cdot \frac{\sin A}{\sin C} \Rightarrow a = 10 \cdot \frac{\sin 30°}{\sin 105°}$$

$$\sin 30° = 1/2$$

$$\sin 105° = \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} + \frac{1}{2} \cdot \frac{1}{\sqrt{2}} = \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

$$\Rightarrow a = 10 \cdot \frac{1/2}{(\sqrt{3}+1)(2\sqrt{2})} = 10 \cdot \frac{1}{2} \cdot \frac{2\sqrt{2}}{\sqrt{3}+1}.$$

$$a = 10 \cdot \frac{\sqrt{2}}{\sqrt{3}+1}.$$

by rationalizing the dinominator $\frac{\sqrt{3}-1}{\sqrt{3}-1}$.

we get,

$$a = 10\frac{\sqrt{2}(\sqrt{3}-1)}{3-1} = 10 \cdot \frac{\sqrt{6}-\sqrt{2}}{2}$$

$$a = 5(\sqrt{6}-\sqrt{2}) \text{ cm}.$$

Q49. If $a = 8$, $b = 6$ and angle $C = 90°$ find c.

Soln

~~a^2~~. given, $a = 8$; $b = 6$; $C = 90°$

~~By Pythogoreas theorem~~,

$\cancel{c^2 = a^2 + b^2}$

~~=>~~ By cosine rule.

$$c^2 = a^2 + b^2 - 2ab\cos C.$$

$$\Rightarrow c^2 = 8^2 + 6^2 - 2(8)(6)\cos 90°$$

$$\Rightarrow c^2 = 64 + 36 - 0.$$

$$\Rightarrow c^2 = 100$$

$$\Rightarrow c = \sqrt{100} = 10.$$

Q50. Prove that, $a/\sin A = b/\sin B = c/\sin C$.

Soln

we know.

$$\Delta = \frac{1}{2} \times \text{base} \times \text{height}.$$

$$\Rightarrow \Delta = \frac{1}{2} c \cdot h_c.$$

$$\Rightarrow \Delta = \frac{1}{2} c (b \sin A) \quad \left[\begin{array}{l} \sin A = \frac{h_c}{b} \Rightarrow h_c = b \sin A \\ \sin B = \frac{h_c}{a} \Rightarrow h_c = a \sin A \end{array}\right]$$

$\Rightarrow \Delta = \frac{1}{2} bc \sin A$ —— (i)

Similarly, $\Delta = \frac{1}{2} ac \sin B$ —— (ii)

$\Delta = \frac{1}{2} ab \sin C$ —— (iii)

$\therefore$ Comparing Eq (i) (ii) & (iii).

$$\frac{1}{2} bc \sin A = \frac{1}{2} ac \sin B = \frac{1}{2} ab \sin C$$

Multiplying the entire eqn by $\frac{2}{abc}$

$$= \frac{bc \sin A}{abc} = \frac{ac \sin B}{abc} = \frac{ab \sin C}{abc}$$

$$= \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\Rightarrow \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

—— Hence proved.

AREA OF CURVILINEAR FIGURE

Q51. State Simpson's one-third rule.

Soln

Simpsons one-third rule is.

$$\frac{h}{3}\left[F + L + 2O + 4E\right]$$

where, $h \Rightarrow$ width of the strip

$F \Rightarrow$ First ordinate

$L \Rightarrow$ Last ordinate

$2O \Rightarrow 2$ (Sum of odd ordinates)

$4E \Rightarrow 4$ (Sum of even ordinates)

Q52. Explain how the area under a curve is estimated using equal ordinates.

Soln

To find the area under a curve the base is divided into equal parts, and ordinates (vertical lines) are drawn at these points. These ordinates form many thin strips. The area of each strip is approximately.

the area of ~~each stri~~ a rectangle. Then all the rectangular areas ~~um~~ are added to get an approximate total area under the curve. As the number of strips increase, the approximation becomes more accurate.

Q53. Using the ordinates 2, 4, 6, 8, 6, 4, 2 over the interval $x = 0$ to $x = 6$ find the area by simpson's rule.

Soln

Given,

$x = 0$ to 6.

$y_0 \cdots y_6 = 2, 4, 6, 8, 6, 4, 2$

Number of sub interval $n = 6$.

$$h = \frac{6-0}{6} = 1.$$

By Simpson's formula ->

$$\frac{1}{3}\left[2 + 2 + 2(2+6+6+2) + 4(4+8+4+2)\right]$$

$$= \frac{1}{3}\left[2+2+2(16) + 4(18)\right]$$

$$= \frac{1}{3}\left[4 + 32 + 72\right]$$

$$= \frac{1}{3}(108) = \frac{108}{3} = 36\ cm^2.$$

SURFACE AREA AND VOLUME

Q55. Write the formula for the curved surface area of a sphere or radius r.

Soln

The formula for the Curved Surface area of a sphere of radius r is. $= 4\pi r^2$

Q56. Define the frustrum of a cone.

Soln

A frustrum of a cone is the part of a right circular cone that remains after a smaller right circular cone is cut off by a plane parallel to the base of the original cone.

Q57. Find the volume of a cone of height 9 cm and radius 4 cm.

Soln

Given, $h = 9\,cm$

$r = 4\,cm$

$$V = \frac{1}{3}\pi r^2 h$$

$$V = \frac{1}{3}\pi (4)^2 (9)$$

$$= \frac{1}{\cancel{3}}\pi \; 16 \times \cancel{9}\,3$$

$$= \pi \; 16 \times 3 \quad = \quad 48\pi\ cm^3$$

Q58. A cylinder of radius 5 cm and height 12 cm is melted to form small spheres of radius 1 cm. How many ~~sphe~~ spheres?

Sol^n Given, $R = 5 cm$
$H = 12 cm$.

Volume of cylinder $= \pi R^2 H$
$= \pi (5)^2 12 = \pi (25)(12)$
$= 300\pi \text{ cm}^3$

Radius of the sphere $= 1cm$

$\therefore$ Volume of Sphere $= \frac{4}{3}\pi r^3$
$= \frac{4}{3}\pi (1)^3 = \frac{4}{3}\pi \text{ cm}^3$

$\therefore$ Number of spheres.

$N = \frac{\text{Volume of cylinder}}{\text{Volume of one sphere}} = \frac{300\cancel{\pi}}{\frac{4}{3}\cancel{\pi}}$

$\Rightarrow \overset{75}{\cancel{300}} \times \frac{3}{\cancel{4}} = 75 \times 3 = 225.$

$\therefore$ 225 small spheres can be formed.

Q 59. Find the total surface area and volume of a frustrum of a cone with radii 6 cm & 4 cm and height 10 cm.

Sol^n Given,

radii = 6 cm & 4 cm.
height = 10 cm.

$l = \sqrt{h^2 + (R-r)^2}$

$= \sqrt{10^2 + (6-4)^2} = \sqrt{100 + 2^2} = \sqrt{100+4}$
$= \sqrt{104}$ cm

$\therefore$ CSA $= \pi (R + r) l$
$= \pi (6+4)\sqrt{104} = 10\pi\sqrt{104} \text{ cm}^2$

Ar. of Base $= \pi R^2 + \pi r^2 = \pi(6^2) + \pi(4^2) = 36\pi + 16\pi$
$= 52\pi \text{ cm}^2$

$\therefore$ TSA = CSA + Are of Base

$= 10\pi\sqrt{104} + 52\pi$

$= 101.98\pi + 52\pi = 153.98\pi$.

Now, volume $= \frac{1}{3}\pi h\,(R^2 + r^2 + Rr)$

$= \frac{1}{3}\pi\,(10)\,(6^2 + 4^2 + (6)(4))$

$= \frac{10\pi}{3}\,(36 + 16 + 24)$

$= \frac{10\pi}{3}\,(76) = \frac{760\pi}{3}$ cm3.

STRAIGHT LINE

Q60. State the distance formula between two points in the plane.

Soln The distance formula between two points in the plane is.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Q61. write the slop intercept form of a straight line.

Soln The slop intercept form of a straight line is $[y = mx + c]$ where,

$m \Rightarrow$ slope

$c \Rightarrow$ y-intercept.

Q62. Find the equation of a line passing through (1,3) with slope 2.

Soln

Given,

$x_1, y_1 = 1, 3$

$m = 2$.

$\therefore\ y - y_1 = m(x - x_1)$

$\Rightarrow y - 3 = 2(x-1)$

$\Rightarrow y - 3 = 2x - 2$

$\Rightarrow y = 2x - 2 + 3$

$\Rightarrow y = 2x + 1$

Ans

Q68. Find the equation of the line perpendicular to $3x + 4y - 5 = 0$ passing through $(2,1)$.

Soln

Given.

$3x + 4y - 5 = 0$.

points $= (2, 1)$

$\therefore y = mx + c$

$\Rightarrow 4y = -3x + 5$

$\Rightarrow y = \frac{-3x}{4} + \frac{5}{4}$ $\therefore m_1 = -3/4$.

$m_2 = \frac{-1}{m_1} = -\frac{1}{-3/4} = 4/3$

$y - y_1 = m_2(x - x_1)$

$y - 1 = \frac{4}{3}(x-2)$

$3(y-1) = 4(x-2)$

$3y - 3 = 4x - 8$

$0 = 4x - 3y - 8 + 3$

$4x - 3y - 5 = 0$.

Ans

Q64. Determine the angle between the lines $2x - y = 0$ and $x + y = 0$

Sol^n

given,

Eq's $\Rightarrow 2x - y = 0$

$\Rightarrow x + y = 0$.

$m_1 \Rightarrow \cancel{-}y = \cancel{-}2x = 2$.

$m_2 \Rightarrow y = -x = -1$

$$\therefore \tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$$

$$= \left|\frac{-1 - 2}{1 + (2)(-1)}\right|$$

$$= \left|\frac{-3}{1 - 2}\right|$$

$$= \left|\frac{-3}{-1}\right| = |3| = 3.$$

$$\theta = \tan^{-1}(3). \approx 71.57°.$$